

Registration Number: _____

Name: _____

Branch & Section: _____

SRM UNIVERSITY – AP, ANDHRA PRADESH

Mid Term Examinations, March 2025

[Question Paper ID: 008244]

Subject : DATA STRUCTURES

Title

Batch : 2024,2023

Degree : B.Tech., M.Tech. Integrated

Branch : CSE

Subject : CSE 102

Code

Max Marks : 25

Duration : 1 hour

QP Set : ---

Instructions:

Read the instructions provided for every question properly before attempting the answer.

PART-A (10 × 1 Marks = 10 Marks)

Answer **all** Questions

Marks BL CO

- | | | | | |
|--|---|---|---|---|
| 1. Consider the following C code:
sum=0;
for(i=1;i<=n;i*=2)
for(j=1;j<=n;j++)
sum++;
The time and space complexities of the above C code are _____ and _____. | 1 | 1 | 2 | 1 |
| 2. Consider the following sequence of operations on an empty stack:
push(54); push(52); pop(); push(55); push(62); s = pop();
Consider the following sequence of operations on an empty queue:
enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q =
dequeue();
The value of s + q is _____. | 1 | 1 | 3 | 1 |
| 3. _____ is the limitation of the Linear Queue. | 1 | 1 | 1 | 1 |
| 4. The overflow condition in circular queue is represented by
a. if((rear+1)%maxsize)==front)
b. if((rear+1)%maxsize)=front)
c. if((rear)%maxsize)==front)
d. None of these | 1 | 1 | 1 | 1 |
| 5. What is the time complexity of a push operation in a stack and to get minimum element from the stack, respectively?
a. O(1) and O(n)
b. O(log n) and O(n)
c. O(n) and O(1)
d. O(1) and O(n log n) | 1 | 1 | 2 | 1 |
| 6. Consider an implementation of unsorted singly linked list. Suppose it has its representation with a head pointer only. Given the representation, which of the following operation can be implemented in O(1) time?
I) Insertion at the beginning of the linked list
II) Insertion at the end of the linked list
III) Deletion of the beginning node of the linked list
IV) Deletion of the last node of the linked list
a. I and II
b. I and III
c. I, II and III
d. I, II and IV | 1 | 1 | 2 | 2 |
| 7. _____ is the time complexity to insert an element at the end of a doubly linked list. | 1 | 1 | 1 | 2 |

8. Which of the following is a **valid** node creation of doubly linked list? 1 1 1 2
- a. struct node {
 struct node prev;
 int data;
 struct node next;
};
- b. struct node* {
 struct node * prev;
 int* data;
 struct node * next;
};
- c. struct node {
 struct node * prev;
 int data;
 struct node * next;
};
- d. struct node {
 struct node * prev;
 struct node * data;
 struct node * next;
};
9. Consider the function f defined below: 1 1 2 2
- ```

struct item
{
 int data;
 struct item * next;
};
int f(struct item *p)
{
 return (
 (p == NULL) ||
 (p->next == NULL) ||
 ((p->data <= p->next->data) && f(p->next))
);
}

```
- For a given linked list p, the function f returns 1 if and only if
- a. Not all elements in the list have the same data value
- b. The elements in the list are sorted in non-decreasing order of data value
- c. The elements in the list are sorted in non-increasing order of data value
- d. None of them
10. In a circular linked list, \_\_\_\_\_ number of nodes are containing NULL as the address. 1      1      1      2

**PART-B (3 × 5 Marks = 15 Marks)**

**Marks BL      CO**

Answer **any 3** Questions

11. Write pseudocode to check for balanced parentheses in an expression using stack operations and describe the methodology in detail. Consider the following examples: 5      5      3      1
- | Example 1                           | Example 2                            |
|-------------------------------------|--------------------------------------|
| Input: s = “[{()}]”<br>Output: true | Input: s = “[{()}]”<br>Output: false |
12. Given the following sequence of letters and asterisks: EAS\*Y\*QUE\*\*\*\*ST\*\*\*IO\*N\*\*\*\* 5      5      3      1
- (a) Consider the queue data structure, supporting two operations insert and remove. Suppose that for the above sequence, each letter (such as E) corresponds to an insert of that letter into the queue and each asterisk (\*) corresponds a remove operation on the queue. Show the sequence of values returned by the remove operations.
- (b) Write the pseudocodes of the operations: insertion and removal that are performed on the queue in the above example. **2+3**
13. (a) What are the disadvantages of a singly linked list? How can they be overcome? 5      5      2      2
- (b) Write a pseudocode that modifies a singly linked list by moving the last element to the front. **2+3**

14.

A single circular linked list is used in a music playlist application, where each song is stored as a node. The playlist follows a continuous loop, allowing the user to navigate forward without reaching the end.

**5      5      3      2**

Task-1: Explain why a circular linked list is a better choice for implementing a music playlist than a singly linked list.

Task-2: Write a pseudocode to insert a new song after a given song in a circular linked list. **2+3**

\* \* \* \* \*